



Transient Ischemic Attack

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A transient ischemic attack (TIA) is focal brain ischemia that causes sudden, transient neurologic deficits and is not accompanied by permanent brain infarction (eg, negative results on diffusion-weighted MRI). Diagnosis is clinical. Carotid endarterectomy or stenting, antiplatelet medications, and anticoagulants decrease risk of stroke after certain types of TIA.

TIA is similar to [ischemic stroke](#) except that symptoms usually last < 1 hour; most TIAs last < 5 minutes. Infarction is very unlikely if deficits resolve within 1 hour. As shown by diffusion-weighted MRI and other studies, deficits that resolve spontaneously between 1 to 24 hours are often accompanied by infarction and are thus no longer considered TIAs.

TIAs are most common among middle-aged and older adults. TIAs markedly increase risk of stroke, beginning in the first 24 hours.

Etiology of TIA

Risk factors for TIA are the same as those for ischemic stroke.

Modifiable risk factors include the following:

- [Hypertension](#)
- Cigarette smoking
- [Dyslipidemia](#)
- [Diabetes](#)
- Insulin resistance
- [Abdominal obesity](#)
- [Obstructive sleep apnea](#)
- Excess alcohol consumption
- Lack of physical activity

- High-risk diet (eg, high in saturated fats, trans fats, and calories)
- Psychosocial stress (eg, [depression](#))
- Heart disorders (particularly disorders that predispose to emboli, such as [acute myocardial infarction](#), [infective endocarditis](#), [valvular heart disease](#), and [atrial fibrillation](#))
- Carotid artery stenosis
- Use of certain drugs (eg, [cocaine](#), amphetamines)
- [Hypercoagulability](#)
- [Vasculitis](#)
- Exogenous estrogen

Unmodifiable risk factors include the following:

- Prior stroke
- Older age
- Family history of stroke
- Male sex

Most TIAs are caused by emboli, usually from carotid or vertebral arteries, although most [causes of ischemic stroke](#) can also result in TIAs.

Uncommonly, TIAs result from impaired perfusion due to severe hypoxemia, reduced oxygen-carrying capacity of blood (eg, profound [anemia](#), [carbon monoxide poisoning](#)), or increased blood viscosity (eg, severe [polycythemia](#)), particularly in brain arteries with preexisting stenosis. Systemic hypotension does not usually cause cerebral ischemia unless it is severe or arterial stenosis preexists because autoregulation maintains brain blood flow at near-normal levels over a wide range of systemic blood pressures.

In **subclavian steal syndrome**, a subclavian artery stenosed proximal to the origin of the vertebral artery “steals” blood from the vertebral artery (in which blood flow reverses) to supply the arm during exertion, causing signs of vertebrobasilar ischemia.

Occasionally, TIAs occur in children with a severe cardiovascular disorder that produces emboli or with a very high hematocrit due to chronic hypoxemia.

Symptoms and Signs of TIA

Neurologic deficits are similar to those of strokes (see table [Selected Stroke Syndromes](#)). Transient monocular blindness (amaurosis fugax), which usually lasts < 5 minutes, may occur when the ophthalmic artery is affected.

TABLE

Selected Stroke Syndromes

Symptoms and Signs	Syndrome
Contralateral hemiparesis (maximal in the leg), urinary incontinence, apathy, confusion, poor judgment, mutism, grasp reflex, gait apraxia	Anterior cerebral artery (uncommon)
Contralateral hemiparesis (worse in the arm and face than in the leg), dysarthria, hemianesthesia, contralateral homonymous hemianopia, aphasia (if the dominant hemisphere is affected) or apraxia and sensory neglect (if the nondominant hemisphere is affected)	Middle cerebral artery (common)
Contralateral homonymous hemianopia, unilateral cortical blindness, memory loss, unilateral 3rd cranial nerve palsy, hemiballismus	Posterior cerebral artery
Monocular loss of vision (amaurosis)	Ophthalmic artery (a branch of the internal carotid artery)
Unilateral or bilateral cranial nerve deficits (eg, nystagmus, vertigo, dysphagia, dysarthria, diplopia, blindness), truncal or limb ataxia, spastic paresis, crossed sensory and motor deficits*, impaired consciousness, coma, death (if basilar artery occlusion is complete), tachycardia, labile blood pressure	Vertebrobasilar system
Absence of cortical deficits plus one of the following: • Pure motor hemiparesis • Pure sensory hemianesthesia • Ataxic hemiparesis • Dysarthria-clumsy hand syndrome	Lacunar infarcts
* Ipsilateral facial sensory loss or motor weakness with contralateral body hemianesthesia or hemiparesis indicates a lesion at the pons or medulla.	

Symptoms of TIAs begin suddenly, usually last 2 to 30 minutes, then resolve completely. Patients may have several TIAs daily or only 2 or 3 over several years. Symptoms are usually similar in successive

carotid attacks but vary somewhat in successive vertebrobasilar attacks.

Diagnosis of TIA

- Resolution of stroke-like symptoms within 1 hour
- Neuroimaging
- Evaluation to identify the cause

Transient ischemic attack (TIA) is diagnosed retrospectively when sudden neurologic deficits referable to ischemia in an arterial territory resolve within 1 hour.

Isolated peripheral facial nerve palsy, loss of consciousness, or impaired consciousness does not suggest TIA. TIAs must be distinguished from other causes of similar symptoms, such as

- [Hypoglycemia](#)
- [Migraine aura](#)
- Postictal [Todd] paralysis (a transient neurologic deficit, usually weakness, of the limb contralateral to the seizure focus)

Because an infarct, a small hemorrhage, and even a mass lesion cannot be excluded clinically, neuroimaging is required. Usually, CT is the study most likely to be immediately available. However, CT may not identify infarcts for > 24 hours. MRI usually detects evolving infarction within hours. Diffusion-weighted MRI is the most accurate imaging test to rule out an infarct in patients with presumed TIA but is not always available.

The causes of a TIA are the same as [causes of ischemic strokes](#); evaluation includes tests for carotid stenosis, cardiac sources of emboli, [atrial fibrillation](#), and hematologic abnormalities and screening for stroke risk factors. Because risk of subsequent ischemic stroke is high and immediate, evaluation proceeds rapidly, usually on an inpatient basis. It is not clear which patients, if any, can be safely discharged from the emergency department. Risk of stroke after TIA or minor stroke is highest within the first 24 to 48 hours, so if either is suspected, patients are typically admitted to the hospital for telemetry and evaluation.

The ABCD2 score is used to estimate risk of stroke after TIA and is calculated by adding the following:

- A (age): $\geq 60 = 1$
- B (blood pressure): Systolic blood pressure ≥ 140 and/or diastolic blood pressure $> 90 = 1$
- C (clinical features): Weakness = 2, speech disturbance without weakness = 1
- D (TIA duration): ≥ 60 min = 2, 10 to 59 min = 1, < 10 minutes = 0
- D2 (diabetes) = 1

Patients with TIA are at high risk of stroke if they have an ABCD2 score > 4 .

Risk of stroke within 2 days based on the ABCD2 score is approximately:

- For a score of 6 to 7: 8%

- For a score of 4 to 5: 4%
- For a score of 0 to 3: 1%

All patients who have had a TIA require CT angiography, magnetic resonance angiography (MRA), or diffusion-weighted MRI of the carotid and cerebral circulation.

Treatment of TIA

- Prevention of strokes

Treatment of transient ischemic attacks is aimed at preventing strokes; antiplatelet medications and statins are used. Carotid endarterectomy or arterial angioplasty plus stenting can be useful for some patients, particularly those who have no neurologic deficits but who are at high risk of stroke (> 70% ipsilateral carotid stenosis). Anticoagulation is indicated if cardiac sources of emboli are present.

Modifying stroke risk factors, when possible, may prevent stroke.

Key Points

- A focal neurologic deficit that resolves within 1 hour is almost always a transient ischemic attack.
- Test as for ischemic stroke.
- Use the same treatments used for secondary prevention of ischemic stroke (eg, antiplatelet medications, statins, sometimes carotid endarterectomy or arterial angioplasty plus stenting).

Drug Information for the Topic



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